Arista Access Node WFD-ACC-48 QoS Congestion Event SOP

Node: WFD-ACC-48 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_2001

Description

This Standard Operating Procedure (SOP) addresses the root cause of QoS congestion events (microbursts, interface tail drops, QoS queue starvation) on the Arista access node WFD-ACC-48, located at the Whitefield POP. The node's role is to provide access connectivity to downstream networks. QoS congestion events occur when the node's egress queues become overwhelmed, leading to packet drops and subsequent network performance degradation. This is typically caused by misconfigured or inadequate QoS policies, which fail to prioritize critical traffic or allocate sufficient buffer resources. Furthermore, the presence of microbursts, wherein a large number of packets are transmitted in a short time frame, can exacerbate the issue by causing interface tail drops. To mitigate this, it is essential to ensure that QoS policies are properly configured, with sufficient buffer resources allocated to egress queues. Additionally, monitoring network traffic and adjusting QoS policies as needed will help prevent congestion events.

Pre-requisite Checks

- Check: Verify redundant link status on WFD-ACC-48 to ensure network connectivity is maintained in case of a primary link failure, Action: Dispatch remote hands to Whitefield POP to perform the link check

- Check: Confirm that the Arista access node WFD-ACC-48 is properly configured for QoS, including egress queue buffer sizes and policing policies, Action: Review WFD-ACC-48 configuration to ensure QoS policies are properly set up

Potential Root Causes

Cause	Probability
Inadequate buffer resources allocated to egress queues	High
Misconfigured or inadequate QoS policies	Medium
Presence of microbursts on egress interface	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces egress, Expected Output: Interface
statistics, including errors and drops, Description: This command will
display detailed statistics for the egress interface, including packet
drops and errors, which can help identify potential congestion issues.

Command: show qos policer, Expected Output: Policer configuration,
including buffer sizes and rates, Description: This command will
display the policer configuration, including buffer sizes and rates,
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```
which can help identify misconfigured or inadequate QoS policies.

Command: show buffer, Expected Output: Buffer usage statistics, including egress queue buffer sizes, Description: This command will display buffer usage statistics, including egress queue buffer sizes, which can help determine if buffer resources are adequate.
```

Resolution Steps

Step 1: Step: Increase egress queue buffer sizes, Command: queue egress buffer-size , Description: This command will increase the buffer size for the egress interface, which can help alleviate congestion issues.

Step 2: Step: Adjust QoS policies to prioritize critical traffic, Command: queue egress policer buffer rate , Description: This command will adjust the QoS policy to prioritize critical traffic, which can help prevent congestion events.

Step 3: Step: Verify buffer usage and adjust as needed, Command: show buffer, Description: This command will display buffer usage statistics, and adjustments can be made as needed to ensure buffer resources are adequate.

Post-Check Verification

```
Command: show interfaces egress, Expected Output: Reduced packet drops and errors, Description: This command will display detailed statistics for the egress interface, including packet drops and errors, which can help verify the effectiveness of the resolution steps.

Command: show qos policer, Expected Output: Policer configuration updated to prioritize critical traffic, Description: This command will display the policer configuration, which can help verify the effectiveness of the resolution steps.

Command: show buffer, Expected Output: Egress queue buffer sizes updated and buffer usage within acceptable limits, Description: This command will display buffer usage statistics, which can help verify the effectiveness of the resolution steps.
```

Roll-back Procedures

- Step: Restore original policer configuration, Command: queue egress policer buffer rate , Description: This command will restore the original policer configuration, which can help ensure that changes did not cause unintended consequences.

- Step: Restore original egress queue buffer sizes, Command: queue egress buffer-size , Description: This command will restore the original egress queue buffer sizes, which can help ensure that changes did not cause unintended consequences.